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PALEOMAGNETIC EVIDENCE FOR THE PALEOPOSITION OF SUMBA ISLAND, INDONESIA

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Jurassic paleomagnetic directions have been obtained for the sedimentary rocks in Sumba The mean direction is 59 2° in declination and −44 2° in inclination The reliability of the direction is ascertained through thermal demagnetization and the presence of normal and reversed polarities Comparison with the Permian paleomagnetic direction of Timor indicates that Sumba was subjected to a clockwise rotation through 79 4° relative to Timor since the Jurassic When Sumba is restored by a counter-clockwise rotation, the Sumba pole is gradually approaching the Jurassic pole of Australia as well as the Timor Permian pole This implies that, at least until the Jurassic, Sumba and Timor were situated at the Australian continental margin and that Sumba rotated clockwise during or after the Jurassic

1 Introduction

The Banda arc is a double arc of which the inner part is a typical volcanic island arc and the outer part a non-volcanic arc (Fig 1) Because of the complex stratigraphy and structure of this area, conspicuously different interpretations of the evolution of the Indonesian Archipelago have been proposed [1–5] According to Audley-Charles [5] the outer Banda arc represents the deformed margin of the Australian continent which, during the pre-Late Jurassic, included Sumba He suggested on the basis of bathymetric morphology and Mesozoic facies distribution that during the Late Jurassic Sumba became detached from the Australian margin by clockwise rotation associated with spreading in the adjacent Wharton basin of the Indian Ocean

Paleomagnetism offers an independent method for testing the various hypotheses In 1979, during an expedition to Sumba, Sumbawa and Flores, samples were collected for paleomagnetic investigation The purpose of the present paper is to discuss the paleoposition of Sumba based on the results of paleomagnetism

2 Geologic outline

2 1 Sumba

Almost the total area of Sumba Island is covered by coral reefs ranging in age from Miocene to Recent Some old rock bodies are found in the southern Coast Mountains, Fanadaro Mountains and the southeast area of the island The samples for the paleomagnetic study were taken from the Southern Coast Mountains, with the exception of nine samples which came

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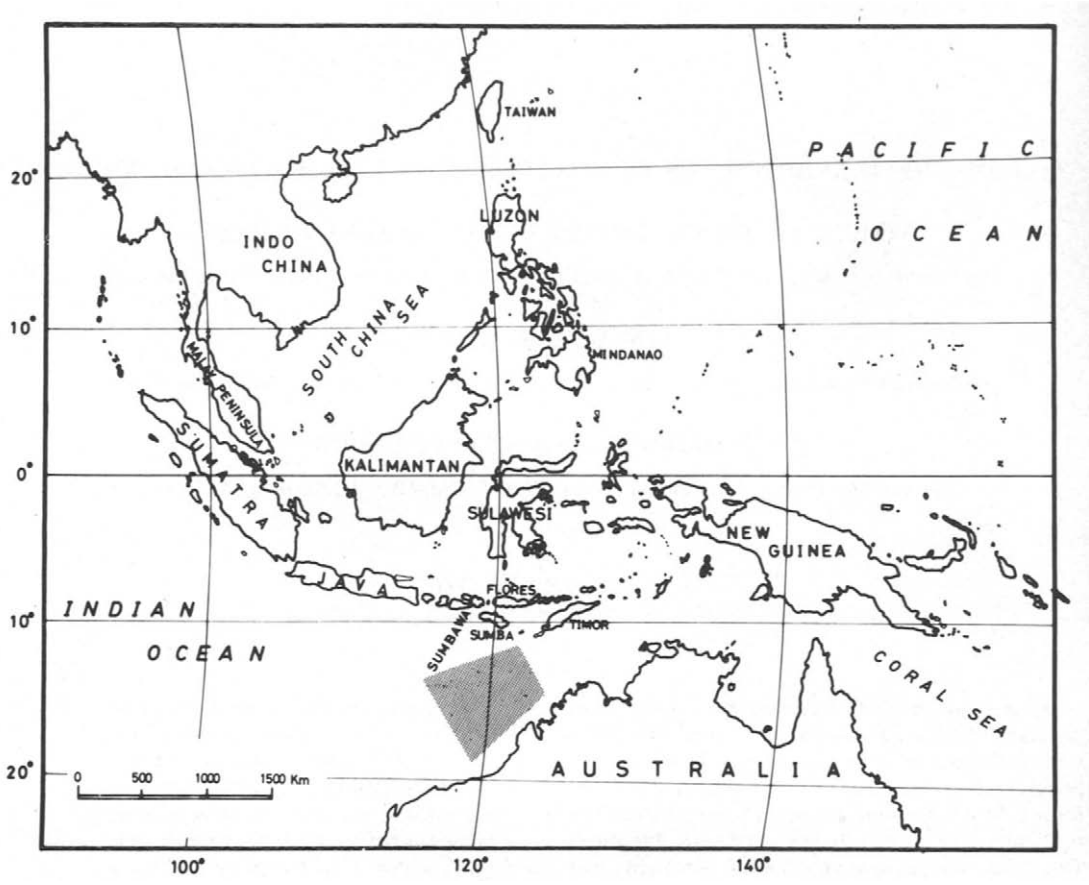


Fig 1 Map showing the configuration of the Indonesian Archipelago with the positions of Sumba, Sumbawa and Flores The shaded area shows the Argo Abyssal Plain where the M-series of marine magnetic anomalies are observed

from the Fanadaro Mountains in the central part of the island In both areas, the strata can be divided into the following four members, pre-Tertiary sedimentary rocks, pre-Tertiary igneous rocks, Paleogene sedimentary rocks and Neogene rocks

The pre-Tertiary sedimentary rocks consist principally of black and hard shales which may be flysch-type sediments These rocks are in-place altered to quartz-sandstone and quartzite at the boundary part of the pre-Tertiary igneous rock body The lamellibranches found in this formation near the south coast of west Sumba were determined as *Inoceramus,* and in the same rock an ammonite was also found [6] The formation was assigned a Jurassic age on the basis of these fossils [6]

The pre-Tertiary igneous rocks are mostly composed of basaltic dykes and granodiorites intruding into the pre-Tertiary sedimentary rock body The age of this rock has not been determined by radiometric methods

The Paleogene sedimentary rocks are composed of mudstones, marls and limestones The age of the formation is considered to be Eocene to Oligocene [7]

The early Neogene rocks are mainly composed of tuffs, tuff breccias, tuffaceous sandstones and alternations of tuffaceous sandstones and black mudstones On the other hand, late Neogene rocks consist of fine chalks grading into marls and water-deposited sediments of volcanic origin

2 2 *Sumbawa and Flores*

In these islands, many volcanoes of Quaternary age are developed in the northern zone of the island arc

TABLE 1

Results of paleomagnetic measurement

Site No	Locality	Estimated age	Rock type	N	AF (Oe)	D (°)	I (°)	α_{95} (°)	k	VGP north pole
Sumba										
IS 135	09°35'S, 119°19'E	Miocene	tuff breccia	8	300	3 0	−20 0	9 2	37 2	87 0°N, 223 5°E
IS 137	09°41'S, 119°25'E	Miocene	tuff	13	300	17 6	16 0	11 1	15 0	65 0°N, 164 5°E
IS 142	09°42'S, 119°27'E	Miocene?	mudstone	4	150	−1 3	−26 9	9 1	103 4	85 3°N, 315 0°E
IS 141	09°45'S, 119°27'E	Tertiary?	dyke	6	300	177 0	35 7	10 6	41 2	79 6°N, 315 2°E
Mean				4		4 6	−17 1	28 4	11 5	85 4°N, 198 0°E
IS 143	09°42'S, 119°27'E	Jurassic	shale	11	150	63 8	−43 1	8 6	29 3	27 7°N, 232 8°E
IS 144	09°42'S, 119°27'E	Jurassic	shale	12	50	59 2	−45 9	4 4	101 0	31 7°N, 235 6°E
IS 146	09°39'S, 119°42'E	Jurassic	shale	9	300	234 5	43 3	20 6	7 2	36 1°N, 234 0°E
Mean				3		59 2	−44 2	5 7	477 6	31 8°N, 234 1°E
Sumbawa										
IS 201	08°39'S, 117°27'E	Miocene	tuff	10	50	−1 3	−25 8	6 9	49 7	84 9°N, 311 8°E
IS 203	08°39'S, 117°26'E	Miocene	lava flow	9	300	−1 8	0 8	4 9	109 4	80 8°N, 106 1°E
Mean				2		−1 7	−12 4			87 1°N, 81 9°E
Flores										
IS 205	08°37'S, 122°24'E	Miocene	lava flow	11	200	−3 9	−19 8	3 5	171 5	85 8°N, 9 7°E
IS 208	08°42'S, [illegible]06'E	Miocene	tuff breccia	8	150	218 4	8 9	6 4	76 3	51 6°N, 208 3°E
IS 209	08°42'S, 122°08'E	Miocene	tuff breccia	8	200	−29 1	−37 5	8 2	46 3	59 4°N, 5 2°E
Mean				3		3 9	−24 6	57 1	5 7	84 3°N, 260 1°E

N = number of specimens, AF = demagnetization field, D = declination, I = inclination, α_{95} = cone of 95% confidence, k = precision parameter

They show a relatively simple structure

In the southern half, some Neogene volcanic and sedimentary rocks are exposed These formations are composed of alternations of volcanic rocks and marine deposits The age of these formations is considered to be Early Miocene

3. Paleomagnetic results

Rocks collected at fifteen sites in Sumba range in age from Jurassic to Miocene Rocks of Jurassic age were hard shales Rocks of Miocene age were obtained from five sites in Flores and four sites in Sumbawa Sampling localities and rock types of each site are listed in Table 1 About ten hand samples were collected in each site A specimen 2 5 cm in diameter and 2 5 cm long was cut from the rock samples in the laboratory

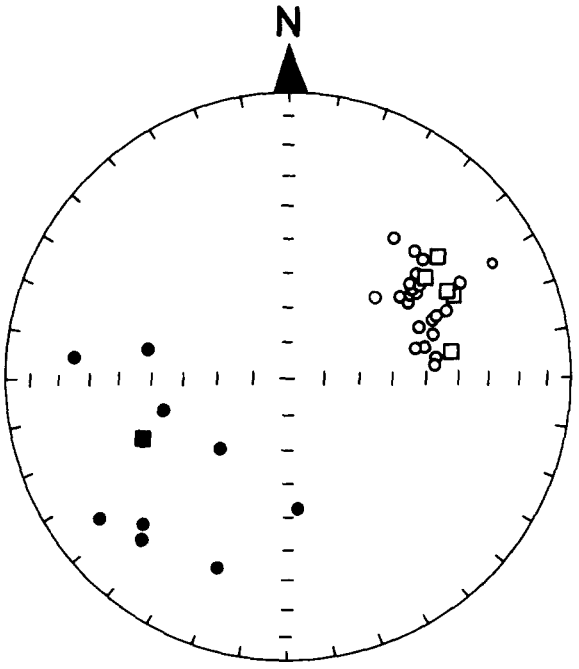


Fig 2 Equal-area projections showing the directions of magnetization of the Jurassic sediments with respect to bedding plane Remanent magnetizations after AF demagnetization at the optimum field intensity and after thermal demagnetization at 200°C are shown with circle and square symbols, respectively Solid symbols on the lower hemisphere, open symbols on the upper hemisphere

The natural remanent magnetization (NRM) was measured on a spinner magnetometer (Schonstedt, SSM-1A) The stability of the NRM was tested by alternating field (AF) demagnetization of three specimens randomly picked from each site Stepwise demagnetization was carried out using a three-axis tumbler with peak fields of 50, 100, 150, 200 and 300 Oe The AF value at which the precision parameter k [8] for three specimens reached a maximum value was adopted as the optimum demagnetization field intensity for the site If the precision parameter k for three specimens was less than 10 0, no further paleomagnetic investigations were conducted for that site

The cleaned directions at the optimum field intensity were grouped and produced a well-defined mean direction The westerly horizontal directions of sample IS 141, which showed a high dispersion, moved toward the reversed present field direction and became much more tightly clustered by demagnetization treatment, i e the parameter k increased from 2 2 to 41.2 Directions of samples IS 143 and IS 144 of Jurassic age shifted upward about 10° and reached a stable end point at the optimum field, while the direction of sample IS 146, which showed reversed polarity, shifted downward by 40° without any change in declination Three site mean directions of Jurassic age clustered nicely after AF demagnetization, i e the parameter k increased from 6 1 to 477 6 Some specimens of Jurassic age were subjected, after AF demagnetization, to thermal demagnetization in steps of 100°C up to 300°C As demonstrated in Fig 2, the thermal treatment caused little changes in the directions This fact suggests that the secondary magnetization is effectively eliminated from the NRM by AF demagnetization at the optimum field intensity

Tilt-corrected remanent directions after AF demagnetization are listed in Table 1, together with their virtual geomagnetic pole (VGP) position Mean directions are also listed for the Miocene of the three islands and for the Jurassic of Sumba

The Miocene rocks of each island have almost the same direction of remanent magnetization to the axial dipole field, although the remanent direction from Flores shows a considerable scattering

One of the Jurassic data reveals a reversed polarity and the mean direction of three Jurassic sites has become tightly clustered after correction for tilting,

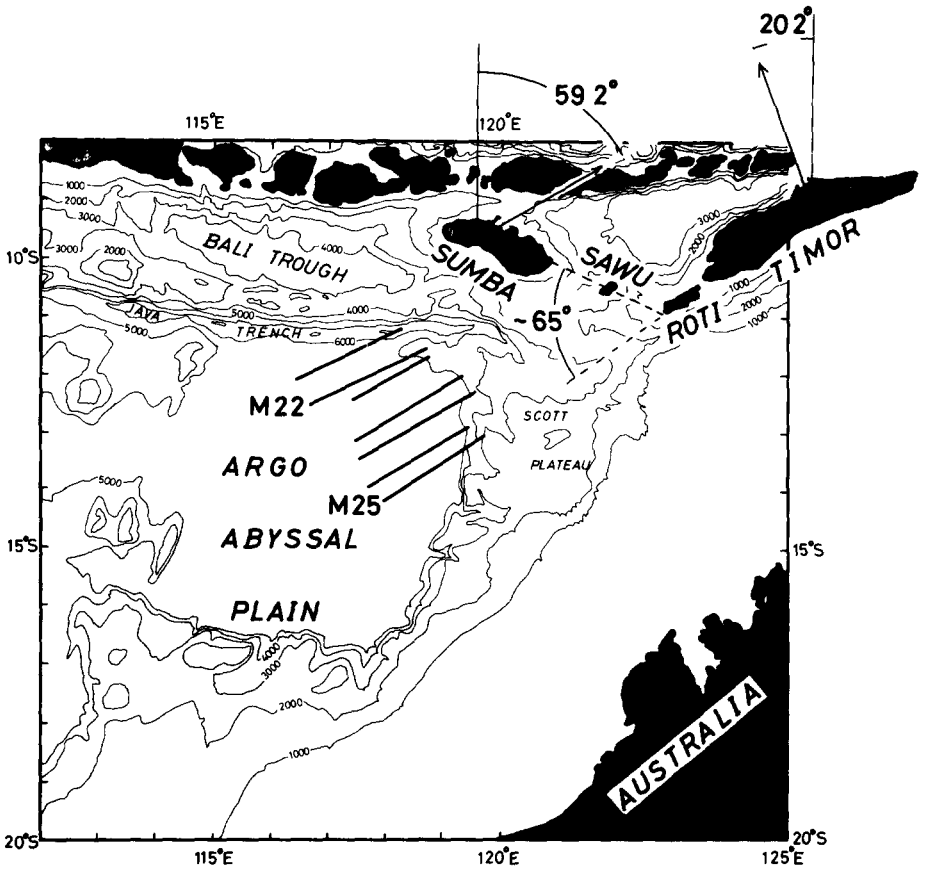


Fig 3 Comparison of the clockwise rotation of Sumba relative to Timor presumed from viewpoints of paleomagnetism, geomagnetism in marine geology and geomorphology Magnetic anomalies in Argo Abyssal Plain from Larson [17] The bending angle of 65° is presumed from geomorphological configuration

i e the parameter k increased from 167 5 to 477 6 These facts suggest a great reliability of the Jarassic paleomagnetic direction

4. Discussion

The mean directions of the Miocene from the three islands are almost the same to the axial dipole field and show considerably similar values These data imply that the islands have been situated at their present position since the Miocene

As shown in Fig 3, the mean declination of the Jurassic rocks of Sumba points to 59 2° eastward of the geographic north The only other paleomagnetic results reported from this region [9] show that the declination of Timor in the Late Permian is 20.2° westward of the geographic north An apparent polar wander has occurred between Late Permian to Early Jurassic time Amounts of polar motion of 14° and 17° are estimated from North America [10–13] and Europe [14–16], respectively The polar motion is, however, seen to cause little change in declination in Sumba and Timor because only the northward shift occurs during polar motion between the Late Permian and the Early Jurassic The Jurassic declination of Sumba is allowable to be compared with the Permian declination of Timor.

Compared with these directions, the clockwise rotation of Sumba is expected to have occurred by 79 4° relative to Timor held fixed From the viewpoint of geomorphological configuration, the trend

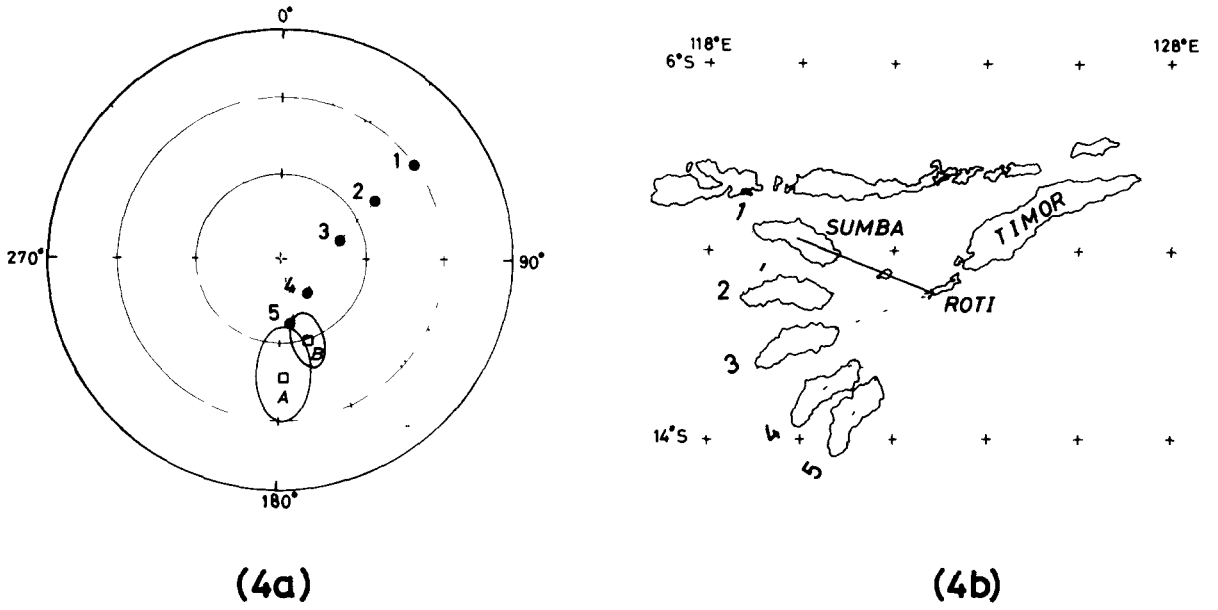


Fig 4. Comparison of pole positions from Sumba, Timor and Australia *A* and *B* correspond to the Early Jurassic pole from Australia and the Late Permian pole from Timor, respectively Poles from Sumba are drawn by filled circles in (a) The numbered poles correspond to the pole evaluated for the rotated position of Sumba with the same number in (b) The present position of Sumba is drawn by configuration *1* in (b)

of the Sumba and Sawu island chain seems to have undergone a clockwise bending of about 65° relative to the trend of the Timor and Roti island chain (see

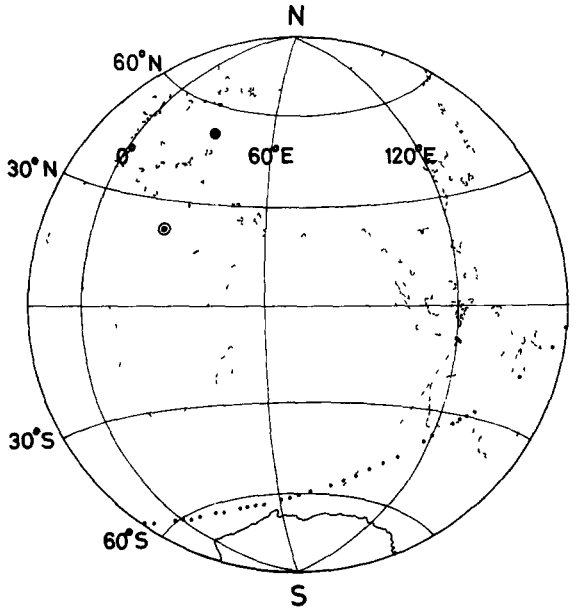


Fig 3) A fair agreement between the geomorphological and paleomagnetic views lends strong support to the hypothesis that Sumba rotated clockwise through roughly 70° about a pole of rotation just west of Roti island since the Jurassic The discrepancy of 14 4° may be attributed to the polar motion between the Late Permian and the Early Jurassic The clockwise rotation of Sumba in the Jurassic is consistent with the pattern of the magnetic anomaly in the Argo Abyssal Plain to the southwest of Sumba (see Fig 3) Larson [17] identified the Argo set as part of the M-series of anomalies which have a basement age of 150 m y and Falvey and Veevers [18] pointed out that the pattern of anomalies indicated a pole of rotation in the vicinity of east Timor

Fig 5 Probable paleoposition of Sumba relative to Antarctica Sumba is situated at the point of intersection between the paleolatitudinal line (dotted line) and the circular arc through Sumba about the pole of rotation of the Indian Ocean (dashed line) ● = Jurassic geomagnetic pole obtained from Antarctica, ○ = pole of rotation inferred from the magnetic lineation of the southwest Indian Ocean

In Fig 4 are shown the paleomagnetic poles for Australia and Timor together with their error circles Poles *A* and *B* correspond to the mean Jurassic pole (47°S, 176°E) of Australia [19,20] and the Late Permian pole (56 6°S, 159 8°E) of Timor [9], respectively These are far away from the VGP obtained from Sumba which is shown by pole *1* This difference may have been caused by the clockwise rotation of Sumba after the fixation of the Jurassic magnetization In order to evaluate the poles for the restored positions, Sumba was rotated counter-clockwise through 23°, 43°, 65° and 79 4° about a pole of rotation just east of Roti island (cf Fig 4b) The poles from Sumba in Fig 4a are numbered in order of increasing rotational angle It is clearly seen that the poles of Sumba are gradually approaching the poles of Australia and Timor as the rotation progresses The pole of Sumba is not statistically distinguished from both the poles of Australia and Timor when Sumba is rotated through 79 4° (from a paleomagnetic point of view presumed to be the probable angle of rotation for Sumba) It may lead us to the conclusion that Sumba and Timor were situated near the Australian continental margin in the Jurassic, and that during or since the Jurassic Sumba rotated clockwise to its present curious position This inference accords with an interpretation [5] based on geological evidence

The inclination of the magnetization provides information about the past angular distance between the site and the pole Combining the inclination data with the plate motion deduced from sea-floor spreading, we estimate that the paleoposition of Sumba was situated near Antarctica as well as Australia

The mean inclination of −44 2° of Sumba indicates a paleolatitude of 25 9°S The position of the Jurassic geomagnetic north pole is different from that of the present pole because the polar wandering is observed from the Jurassic to the present [12] A Jurassic pole at 53°N, 35°E is known from Antarctica [21], and by assuming the Antarctic paleomagnetic pole to be the geographic pole, we can trace the 29 5°S paleolatitude on which Sumba was situated in the Jurassic, as shown by a dotted line in Fig 5.

Two sets of marine magnetic anomaly patterns with different periods to the south of Sumba are reported, the M-series in the Argo Abyssal Plain [17] and anomalies 1 to 17 in the Indian Ocean [22] The opening of the Southeast Indian Ocean is regarded as the main cause of drift of Sumba from Antarctica The pole of rotation of the Indian Ocean floor is obtained as 26°N, 21°E by the fracture zone method of Heirtzler et al [23] Sumba could have lain on the locus shown by the dashed line in Fig 5 which represents the path of the restoring rotation of Sumba The paleoposition of Sumba relative to Antarctica is assumed to lie at the point of intersection between the paleolatitudinal line and the locus of the drift (Fig 5)

It is noted that there is a distinct difference between the poles obtained from Antarctica and the Asian continent The Asian Jurassic poles, deduced from the Russian Platform and China, are 65°N, 138°E and 55°N, 149°E, respectively [21] This difference is mainly due to the difference of the longitude values of both poles Both latitudes, on the other hand, have almost the same value This agreement suggests that Antarctica and the Asian continent have exhibited almost the same geographical display on the earth as the present one although a significant rotation had occurred between them around the present spin axis, and that Antarctica has never approached the Asian continent since the Jurassic We, therefore, concluded that during the Jurassic Sumba was situated far away from the Asian continent, at a fairly high latitude in the southern hemisphere of the present geographical coordinate system

5 Concluding remarks

Based on the paleomagnetic study of Sumba, Sumbawa and Flores, together with the marine magnetic anomaly pattern, it is inferred that

(1) Sumba formed a marginal part of the Australian continent occupying a high latitude in the southern hemisphere at least until the Jurassic

(2) Sumba rotated clockwise during or after the Jurassic This conclusion based on paleomagnetic data accords with an interpretation [5] based on geological and bathymetric evidence

(3) Sumba ceased its northward movement and clockwise rotational motion since the Miocene

(4) The three islands, Sumba, Sumbawa and Flores, have been approximately at their present positions without any rotational motion since the Miocene

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